

Q1

Math Crossword is an engaging and educational puzzle game that combines the challenge of crossword puzzles with the excitement of solving math problems. It is like a word crossword puzzle but with a twist i.e. instead of letters, numbers or sums/difference of numbers (either given or to find) need to be justified into the grid to solve the puzzle.

A particular group of 5 cells horizontally or vertically forms an equation. The variables (either operand or result) in the equations are required to be adjusted in such a way that the dependencies of the equations are preserved.

Consider two Math Crossword given below with their solutions in orange color:

|    |   |    |   |   |
|----|---|----|---|---|
| X  | - | 10 | = | X |
| -  | Q | +  | Q | - |
| X  | + | X  | = | 5 |
| =  | Q | =  | Q | = |
| 14 | - | 13 | = | X |

|    |   |    |   |   |
|----|---|----|---|---|
| 16 | - | 10 | = | 6 |
| -  |   | +  |   | - |
| 2  | + | 3  | = | 5 |
| =  |   | =  |   | = |
| 14 | - | 13 | = | 1 |

|   |   |    |   |   |   |   |   |    |
|---|---|----|---|---|---|---|---|----|
| X | + | 3  | = | X | - | 9 | = | X  |
| - | Q | +  | Q | - | Q | - | Q | +  |
| X | - | 4  | = | 5 | - | X | = | 0  |
| = | Q | =  | Q | = | Q | = | Q | =  |
| 4 | + | X  | = | X | - | 4 | = | X  |
| + | Q | +  | Q | + | Q | - | Q | +  |
| 1 | + | X  | = | 8 | + | X | = | 10 |
| = | Q | =  | Q | = | Q | = | Q | =  |
| X | + | 14 | = | X | - | X | = | X  |

|    |   |    |   |    |   |   |   |    |
|----|---|----|---|----|---|---|---|----|
| 13 | + | 3  | = | 16 | - | 9 | = | 7  |
| -  |   | +  |   | -  |   | - |   | +  |
| 9  | - | 4  | = | 5  | - | 5 | = | 0  |
| =  |   | =  |   | =  |   | = |   | =  |
| 4  | + | 7  | = | 11 | - | 4 | = | 7  |
| +  |   | +  |   | +  |   | - |   | +  |
| 1  | + | 7  | = | 8  | + | 2 | = | 10 |
| =  |   | =  |   | =  |   | = |   | =  |
| 5  | + | 14 | = | 19 | - | 2 | = | 17 |

Write a C program to solve 5x5 and 9x9 size Math Crossword using Backtracking strategy to fill the values from a range of [0..20] in a box containing X while ignoring the boxes containing Q. Your code should emphasize on the isSafe function which is also to be explained in your answer sheet along with the time complexity analysis.

Further, extend your program to solve a 13x13 Math Crossword.

**Q2 (a)**

In a Rat Maze Problem with a binary matrix of size  $N * M$ , where 0 represents an empty space while 1 represents a wall you cannot walk through. Find the minimum number of 1's that needs to be changed to 0 to reach from  $(x, y)$  to  $(i, j)$ . When  $x, y, i, j$  are input.

Sample Input:

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 1 | 0 |

1, 1, 7, 7

3 walls to be broken: (1, 1), (1, 2), (5,7)

3,7, 7, 7

1 walls to be broken: (5,7)

OR

**Q2 (b)**

An edge in an undirected connected graph is a bridge if removing it disconnects the graph. For example in the graph with following adjacency matrix the edge 6-5 is a bridge:

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 |

Write a linear complexity algorithm to detect all the bridges in the graph.

Further a node is vulnerable, if its degree is least in the graph. Extend your code to find the vulnerable nodes using Adjacency Lists.